

2026 ACC/AHA PRACTICE GUIDELINE FOR HIGH BLOOD PRESSURE

Classification, Blood Pressure Measurement & Treatment Targets

1. BLOOD PRESSURE CLASSIFICATION CATEGORIES

Blood pressure (BP) must be categorized based on an average of ≥ 2 careful in-office readings obtained on ≥ 2 separate occasions using a validated automated oscillometric device:

? Normal BP: Systolic < 120 mmHg AND Diastolic < 80 mmHg.

? Elevated BP: Systolic 120 to 129 mmHg AND Diastolic < 80 mmHg.

? Stage 1 Hypertension: Systolic 130 to 139 mmHg OR Diastolic 80 to 89 mmHg.

? Stage 2 Hypertension: Systolic ≥ 140 mmHg OR Diastolic ≥ 90 mmHg.

? Hypertensive Crisis: Systolic > 180 mmHg and/or Diastolic > 120 mmHg.

2. OUT-OF-OFFICE BP CONFIRMATION

Ambulatory Blood Pressure Monitoring (ABPM) or standardized Home BP Monitoring (HBPM) is recommended to confirm hypertension and rule out white-coat hypertension before pharmacotherapy initiation.

3. UNIVERSAL BLOOD PRESSURE TARGETS

? Primary Target: BP $< 130/80$ mmHg is recommended for all non-pregnant adult patients with confirmed hypertension.

? High-Risk Populations: Patients with Type 2 Diabetes, Chronic Kidney Disease (CKD), Congestive Heart Failure, or Age ≥ 65 years share the universal target of BP $< 130/80$ mmHg (Class I, Level A).

4. THRESHOLDS FOR PHARMACOTHERAPY INITIATION

? Stage 1 Hypertension: Initiate single-agent pharmacotherapy if estimated 10-year ASCVD risk is $\geq 10\%$, or if patient has known diabetes, CKD, or clinical cardiovascular disease.

? Stage 2 Hypertension: Initiate combination pharmacotherapy immediately with TWO first-line agents of different pharmacological classes.

2026 ACC/AHA PRACTICE GUIDELINE FOR HIGH BLOOD PRESSURE

First-Line Antihypertensive Classes, Regimens & Dosing Protocols

5. FIRST-LINE ANTIHYPERTENSIVE CLASSES

Four primary drug classes have demonstrated equivalence in reducing major cardiovascular events and mortality:

1. ANGIOTENSIN-CONVERTING ENZYME (ACE) INHIBITORS:

? Lisinopril: Initial 10 mg orally once daily; titrate to target 20 to 40 mg once daily.

? Enalapril: Initial 5 mg orally once or twice daily; titrate to 10 to 40 mg daily.

? Key Adverse Effects: Dry cough (5-20% due to bradykinin accumulation), hyperkalemia, angioedema. Contraindicated in pregnancy.

2. ANGIOTENSIN II RECEPTOR BLOCKERS (ARBs):

? Losartan: Initial 50 mg orally once daily; titrate to 100 mg once daily.

? Valsartan: Initial 80 mg orally once daily; titrate to 160 to 320 mg once daily.

? Preferred alternative in patients experiencing ACE inhibitor-induced cough. Contraindicated in pregnancy.

3. CALCIUM CHANNEL BLOCKERS (DIHYDROPYRIDINE CCBs):

? Amlodipine: Initial 5 mg orally once daily; titrate to maximum 10 mg once daily.

? Nifedipine Extended-Release: 30 to 60 mg orally once daily (maximum 90 mg daily).

? First-line choice in Black patients and elderly individuals with isolated systolic hypertension.

? Adverse Effect: Dose-dependent peripheral ankle edema.

4. THIAZIDE / THIAZIDE-LIKE DIURETICS:

? Chlorthalidone: 12.5 to 25 mg orally once daily. Preferred over HCTZ due to prolonged half-life and proven cardiovascular event reduction.

? Hydrochlorothiazide (HCTZ): 25 to 50 mg orally once daily.

? Adverse Effects: Hypokalemia, hyponatremia, hyperuricemia (may precipitate gout).

TABLE 1: FIRST-LINE ANTIHYPERTENSIVE DOSAGES

Lisinopril (ACEi): 10 - 40 mg qd (Check K⁺ and Cr)

Losartan (ARB): 50 - 100 mg qd (Alternative to ACEi)

Amlodipine (CCB): 5 - 10 mg qd (First-line in Black adults)

Chlorthalidone (Diuretic): 12.5 - 25 mg qd (Preferred over HCTZ)

2026 ACC/AHA PRACTICE GUIDELINE FOR HIGH BLOOD PRESSURE

Combination Strategies, Resistant Hypertension & Fourth-Line Therapy

6. COMBINATION THERAPY STRATEGIES

In Stage 2 hypertension (systolic BP ≥ 140 mmHg or $\geq 20/10$ mmHg above goal), initiate prompt treatment with two first-line agents, ideally as a single-pill fixed-dose combination to maximize patient adherence.

RECOMMENDED COMBINATIONS:

- ? ACE Inhibitor + Dihydropyridine CCB (e.g., Lisinopril + Amlodipine).
- ? ARB + Thiazide Diuretic (e.g., Losartan + Chlorthalidone).
- ? CCB + Thiazide Diuretic (e.g., Amlodipine + Chlorthalidone).

CONTRAINDICATED COMBINATION:

Dual blockade of the renin-angiotensin-aldosterone system (combining an ACE inhibitor with an ARB or direct renin inhibitor) is strictly contraindicated due to elevated risks of hyperkalemia, acute kidney injury, and syncope without added cardiovascular benefit.

7. RESISTANT HYPERTENSION DEFINITION

Resistant hypertension is defined as blood pressure that remains above target ($> 130/80$ mmHg) despite concurrent adherence to optimal doses of THREE antihypertensive drug classes, one of which must be a diuretic.

STEP 4 THERAPY: MINERALOCORTICOID RECEPTOR ANTAGONISTS (MRAs)

- ? Spironolactone is the drug of choice for resistant hypertension.
- ? Dosage: 25 to 50 mg orally once daily.
- ? Safety Requirement: Baseline serum potassium must be < 4.5 mEq/L and eGFR must be ≥ 30 mL/min/1.73m². Recheck serum potassium and creatinine within 1 to 2 weeks following initiation.

2026 ACC/AHA PRACTICE GUIDELINE FOR HIGH BLOOD PRESSURE

Hypertensive Crises: Emergencies vs Urgencies & ICU Protocols

8. HYPERTENSIVE CRISIS MANAGEMENT

Hypertensive crisis is defined as severe blood pressure elevation (Systolic > 180 mmHg and/or Diastolic > 120 mmHg).

A. HYPERTENSIVE EMERGENCY:

Severe BP elevation accompanied by acute, progressive target organ damage (e.g., acute coronary syndrome, acute pulmonary edema, acute aortic dissection, hypertensive encephalopathy, acute ischemic stroke, or acute renal failure).

? Setting: Immediate admission to an intensive care unit (ICU).

? Treatment Protocol: Continuous intravenous titratable antihypertensive infusions.

? BP Reduction Goal: Reduce Mean Arterial Pressure (MAP) by no more than 20% to 25% within the first hour to prevent cerebral hypoperfusion, then reduce toward 160/100 mmHg over the next 2 to 6 hours.

PREFERRED INTRAVENOUS EMERGENCY AGENTS:

? Nicardipine IV: 5 mg/h IV infusion, titrate by 2.5 mg/h every 5-15 min to max 15 mg/h.

? Labetalol IV: 20 mg initial IV push over 2 min, then 40-80 mg every 10 min, or continuous infusion 1-2 mg/min.

? Sodium Nitroprusside: 0.3 to 2 mcg/kg/min IV; reserved for acute pulmonary edema or refractory crisis.

B. HYPERTENSIVE URGENCY:

Severe BP elevation (> 180/120 mmHg) WITHOUT acute target organ damage.

? Setting: Outpatient or emergency observation.

? Treatment: Re-initiation or upward titration of oral agents (e.g., oral Captopril, Labetalol, or Amlodipine). Rapid intravenous reduction is contraindicated and hazardous.